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PERFORMANCE SPECIFICATION

PUMPING ASSEMBLIES, WHEEL-MOUNTED, DIESEL ENGINE DRIVEN, BULK TRANSFER, FUEL AND POTABLE WATER PUMPING SERVICE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Abstract. This specification covers wheel-mounted, diesel-engine-driven, self-priming, centrifugal, four-inch and six-inch pumping assemblies for fuel and potable water service.

1.2 Classification. The pumping assemblies will be of the following types, sizes and styles as specified (see 6.2).

Type I - Pumping Assembly, Fuel

Size 1 - 350 gallons per minute (GPM) Capacity at 275 Feet Total Head

Style A -Without regulator

Style B - With speed/pressure regulator

Size 2 - 600 GPM Capacity at 350 Feet Total Head
(1250 GPM Capacity at 180 feet total head)

Type II - Pumping Assembly, Water

Size 1 - 350 GPM Capacity at 275 Feet Total Head

Size 2 - 600 GPM Capacity at 350 Feet Total Head

Style A -Without regulator

Style B -With speed/pressure regulator

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S.Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-D/210, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4320

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-52598	Couplings, Clamp, Pipe; with Bolts and Synthetic-Rubber Gaskets for Grooved-End Pipe

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI Z 535.2	Environmental and Facility Safety Signs

(Applications for copies should be addressed to the American National Standard Institute, 11 W. 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B 1.1	Unified Inch Screw Threads (UN and UNR Thread Form).
ASME B 1.20.1	Pipe Threads, General Purpose (Inch).
ASME B 1.20.3	Dryseal Pipe Threads (Inch) Revision and Redesignation of B2.2-1968

(Applications for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F 1122 Quick Disconnect Couplings

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive Philadelphia PA 19428-2959.)

HYDRAULIC INSTITUTE

ANSI/HI 1.1-1.5 Centrifugal Pumps.

ANSI/HI 1.6 Centrifugal Pump Tests

(Application for copies should be addressed to the Hydraulic Institute, 9 Sylvan Way, Parsippany, NJ 07054-3802.)

MANUFACTURERS STANDARDIZATION SOCIETY (MSS)
OF THE VALVE AND FITTING INDUSTRY INC.

MSS SP-80 Bronze, Gate, Globe, Angle, and Check Valves

(Application for copies should be addressed to the Manufacturers Standardization Society of the Valve And Fitting Industry Inc., 127 Park St. NE, Vienna, VA 22180.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-J513 Refrigeration Tube Fittings

SAE J847 Towing Eyes

SAE J614 Oil Dip Stick

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC.

Yearbook

Supplementary Service Data Book

(Application for copies should be addressed to the Tire and Rim Association, Inc. 3200 W. Market Street, suite 304, Akron, OH.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. The materials selected for construction are the responsibility of the contractor. The materials shall be of sufficient durability to meet all the performance requirements as specified herein. Type I pumping assemblies may be constructed of any materials suitable for fuel pumping

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service. Type II pumping assemblies may be constructed of any materials suitable for potable water pumping service. Lead shall not be used. The finished type II pumping assembly shall contain no materials or substances that might leak or disintegrate and cause the pumped water to become non-potable. The material shall have no adverse effect on the health of personnel when used for its intended purpose. The lines and all surfaces that contact potable water shall conform to the applicable federal regulations for use with potable water.

3.2.1 Deterioration prevention and control. The pumping assemblies shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration to which the material is susceptible.

3.2.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Environmental conditions.

3.3.1 Operating temperature. Pumping assemblies shall perform as specified herein in the following ambient temperatures (see 4.5.2 and 4.5.3).

Type	Size	Low temperature °F	High temperature °F
Type I	Size 1	-25	120
Type I	Size 2	-25	135
Type II	Size 1	32	120
Type II	Size 2	32	135

3.3.2 Storage temperature. Pumping assemblies shall not be damaged by storage in any ambient temperature from -65 °F to 155 °F, except type I, size 2 and type II, size 2 that shall withstand -65 °F to 160 °F (see 4.5.2 and 4.5.3).

3.3.3 Drop. Each pumping assembly shall be capable of withstanding drops onto its pintle and frame, about its axle, without permanent damage, deformation, or performance degradation.

3.3.4 Shock. The pump assemblies in transit condition, and prepared for field use, shall be capable of withstanding the shocks normally induced by loading and unloading and impact at speeds up to 8 miles per hour.

3.3.5 Fording capability. The pumping assemblies shall be capable of fording in water to a depth of not less than 30.00 inches without damage.

3.3.6 Slope capability. Type I, size 1, and type II, size 1, pumping assemblies shall be capable of negotiating side slopes up to 20 percent; and approach, negotiate, and depart a ramp of up to 30 percent slope without contact of any component (except the tires).

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3.3.7 Mobility. Type I, size 1, and type II, size 1, pumping assemblies shall be capable of being towed, without malfunction, permanent deformation, or damage, over roads as follows:

a. Improved (paved) roads at an average speed of 20 miles per hour (mph).
b. Unimproved (unpaved) roads at an average speed of 10 mph.
c. Over cross country terrain at an average speed of 8 mph.

Type I, size 2, and type II, size 2, pumping assemblies shall be capable of being towed, without malfunction, permanent deformation, or damage, over roads as follows:

a. Improved (paved) roads at an average speed of 50 miles per hour (mph).
b. Unimproved (unpaved) roads at an average speed of 20 mph.
c. Over cross country terrain at an average speed of 8 mph.

3.4 Maintainability.

3.4.1 Maintenance ratio. The pumping assembly shall have a maintenance ratio of not greater than 0.07. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components, and scheduled before-and-after-operation checks are excluded.

3.4.2 Ease of maintenance. All major assemblies and installed attachments requiring repair, replacement, and adjustment shall be accessible without the removal of other major components. Covers, plates, and mounting brackets that must be removed for component adjustment, repair, or replacement shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. All drain plugs and filler caps shall be durable and shall be permanently attached to the unit without impeding their removal. Corrosion resistance preventive maintenance shall be limited to routine washing, periodic inspection and repair of accident damage.

3.4.3 Reliability. The specified mean-time-between-failures shall be not less than 200 hours when the pumping assemblies are tested in accordance with 4.5.8.

3.5 Performance.

3.5.1 Hydrostatic pressure. The pumps shall withstand an internal hydrostatic pressure of 200 PSI, at rated speed, without leaking or becoming deformed.

3.5.2 Performance. The pumps shall meet the capacities indicated below using water at standard conditions, as specified by the Hydraulic Institute ANSI/HI 1.1-1.6 Centrifugal Pumps and Tests, and positioned at any direction or angle up to 15 degrees from the horizontal.

a. Type I, size 1 - 350 gpm with a total dynamic head of 275 feet.
b. Type I, size 2 - 600 gpm with a total dynamic head of 350 feet and 1250 gpm with a total dynamic head of 180 feet.
c. Type II, size 1 - 350 gpm with a total dynamic head of 275 feet.
d. Type II, size 2 - 600 gpm with a total dynamic head of 350 feet.

3.5.3 Pump priming. The pumps shall prime themselves automatically, after initial filling of the case, and deliver not less than their rated flows in not greater than 2 minutes, when operating at a

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static suction lift equivalent to 10 feet at sea level and atmospheric conditions of 29.92 inches of mercury, using water with a specific weight of 62.3 pounds per cubic foot at 68 °F.

3.5.4 Pump characteristics and accessories. All pumps shall contain drain plugs large enough to allow complete pump draining in less than one minute. All threaded parts shall conform to ASME B 1.1 or ASME B 1.20.3, as applicable.

3.5.4.1 Type I case and volute. Type I pumping assembly case and volute shall be one piece. The cutwater shall be integrally cast into the pump case. The case shall have a pump drain and a 1.50 inch priming port with a check valve. The priming port shall be equipped with a male, quick disconnect, coupling and dust cap in accordance with ASTM F 1122, standard class.

3.5.4.2 Type II case and volute. Type II pumping assembly case and volute shall be one piece. The cutwater shall be integrally cast into the pump case. The case shall have a pump drain and a 1.50 inch priming port with a check valve. The priming port shall be equipped with a male, quick disconnect, coupling and dust cap in accordance with ASTM F 1122, standard class.

3.5.4.3 Impeller and wear rings. The impeller and wear ring shall be fabricated of inherently compatible materials, in a combination suitable for intended service.

3.5.4.4 Seals. The pump shall be equipped with self-lubricating, mechanical seals that will not be damaged or compromised by operation of the pumping assembly under any condition specified herein.

3.5.4.5 Piping and fittings. All piping, valves, and fittings on the suction and discharge of the pumping assemblies shall be flanged except as specified herein. All piping and fittings shall be mounted securely and shall withstand an internal working pressure of 1.5 times the shutoff pressure, at rated speed without leaking or becoming deformed. The discharge piping shall contain a 0.25 inch National Pipe Thread (NPT) vent valve to remove air during priming. The vent shall be located at the highest point in the discharge piping and the vent valve shall be routed to the exterior of the pump and engine housing for easy accessibility.

3.5.4.6 Type I, size 1, and type II, size 1. Type I, size 1, and type II, size 1, pumping assemblies shall have the following suction and discharge components:

a. Two four-inch gate valves shall control the suction of the pump. The gate valves shall be in accordance with MSS SP-80, except that materials for type I shall be a durable, rigid, lightweight, corrosion resistant, non-magnetic material, suitable to the application and compatible with aluminum. The valves shall be supported by a bracket connected to the pumping unit frame. One of these valves shall be in line with the inlet of the pump. Suction couplers shall be four-inch, female, quick disconnect couplings, gaskets, and dust plugs in accordance with ASTM F 1122, standard class, and shall be slanted downward 30° from the horizontal.

b. The discharge of the pump shall be manifolded into two, four-inch gate valves. The gate valves are to be in accordance with MSS SP-80, except that materials for type I shall be a durable, rigid, lightweight, corrosion resistant, non-magnetic material, suitable to the application and compatible with aluminum. The valves shall be supported by a bracket connected to the pumping unit frame. Discharge couplers shall

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be four-inch, male, quick disconnect couplings, gaskets, and dust caps in accordance with ASTM F 1122, standard class, and shall be slanted downward 30° from the horizontal.

c. A suction strainer assembly, consisting of a 0.25 inch mesh, in-line, reinforced basket strainer of a durable, rigid, corrosion resistant material suitable to the application.

3.5.4.7 Type I, size 2. The following components shall be furnished with this pumping assembly:

a. Six-inch, single grooved, flanged adapters, suction and discharge, that accept couplings conforming to A-A-52598.

b. Discharge check valve, having a six-inch, flanged housing of a durable, rigid, light-weight, corrosion resistant, non-magnetic material, suitable to the application and compatible with aluminum.

c. Six-inch gate valves, discharge and suction. The valves shall be positioned to accept both hose and pipe. The valves shall be supported by a bracket connected to the pumping unit frame.

d. A suction strainer assembly, consisting of a 0.25 inch mesh, in-line type, reinforced basket strainer of a durable, rigid, corrosion resistant material, suitable to the application.

3.5.4.8 Type II, size 2. The following components shall be furnished with this pumping assembly:

a. Six-inch, single grooved, connections, suction and discharge, that accept couplings conforming to A-A-52598.

b. A coupling conforming to A-A-52598 and a blind adapter (closure plate/flange), to prevent entry of foreign material, for both the suction and discharge connections. The coupling and gasket shall conform to the requirements herein for potable water contact.

3.5.5 Engine. All pumping assemblies shall be provided with an industrial diesel engine capable of operating on diesel fuel, DF-A, DF-2, JP-5, or JP-8. The engine shall deliver power at its continuous duty rating to meet the pump performance requirements of 3.5.2. The engine shall be equipped with, but not limited to, the following items.

a. A 24-volt electric starting system; except type I, size 1, and type II, size 1 that are 12 volt, with charging alternator and batteries suitable for -25 °F starting; except type II that is 32 °F starting.

b. A heavy duty, dry type, air cleaner with replaceable cartridge and restriction indicator.

c. Fuel filters; primary, water separating type; secondary, each with provision for drainage into a container.

d. Manual engine controls to include a waterproof push-button starter switch and engine run/stop switch.

e. Automatic engine shutdown controls in the event of engine overspeed, dieseling, high engine temperature, or low lubricant oil pressure. All controls shall have manual resets.

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f. Exhaust system with guard and rain cap exiting through the top of the housing. Muffler must have a provision for draining combustion produced water to the exterior of the pumping assembly below the housing.

g. Engine oil drain tube, easily accessible through the outer housing.

h. An auxiliary charging receptacle easily accessible through the outer housing for recharging the batteries from a remote power source without operation of the engine (see figure 1).

i. Engine oil sample valve allowing easy access to the pressure side of the system.

j. Instrumentation (for location on the control and instrument panel), consisting of lubricating oil pressure gauge, tachometer-hourmeter, battery charging indicator, air cleaner restriction indicator, and for type I, size 2, and type II, size 2, an engine temperature gauge.

k. Heavy duty cooling system capable of maintaining the manufacturers recommended operating temperature when performing at rated capacity and temperatures as listed herein. If liquid cooled, the radiator shall have provisions for easy cleaning with compressed air.

l. Engine fan belt guard providing easy access and requiring no tools for removal.

m. Oil dip stick conforming to SAE J614.

3.5.6 Control and instrument panel. The control and instrument panel shall be covered by a latching door with automatically engaging hold-open devices. The panel shall be illuminated for night operation and shall include a dimmer switch that allows adjustment from completely off to full illumination. Lights shall not extend more than one inch horizontally from the vertical face of the panel. The panel shall contain a switch that automatically turns off the lights when the panel door is closed. The control and instrument panel shall include the following as a minimum.

a. Helical type Bourdon tube suction gauge.

b. Helical type Bourdon tube pressure gauge.

c. Tachometer-hourmeter.

d. Voltmeter; for type 1, size 1, and type II size 1, use ammeter.

e. Engine oil pressure gauge.

f. Engine low oil pressure shutdown indicator with manual reset.

g. Engine overspeed shutdown indicator with manual reset.

h. Engine overheated shutdown indicator with manual reset.

i. Engine temperature gauge, for type I, size 1, and type II, size 1, only.

j. A multi-position switch indicating stop, run, and start, at a minimum, for type I, size 1, and type II, size 1, only.

k. Air filter restriction indicator.

l. Manual throttle.

m. Emergency stop button, for type I, size 1, and type II, size 1, only.

3.5.6.1 Wiring. All battery and starting system connections shall be equipped with rubber boots. The positive battery cable shall be red and its terminal labeled with a “+”. The negative

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battery cable shall be black and its terminal labeled with a "-". All wiring shall be gathered, routed, and securely clamped to prevent excessive movement and inadvertent damage during maintenance procedures. Grounding of all electrical components shall be done at a single point.

3.5.7 Automatic pump controller. Type I, size 1, style B, and type II, size 2, style B pumping assemblies shall be equipped with automatic pump controllers which reduce engine speed to limit high discharge pressure or low suction pressure. The controller shall sense the discharge and at 150 psi begin to reduce engine speed. If discharge pressure continues to increase, the controller shall continue to reduce engine speed until the idle setting is reached. The controller shall also sense the pressure on the suction side of the pump. If inlet pressure falls below 20 psi the controller shall begin reducing engine speed. If inlet pressure continues to fall, when pressure is reduced to 10 psi engine speed shall be reduced to idle. The controller shall be capable of being turned off so that engine speed is regulated by the engine governor via the hand throttle.

3.5.7.1 Operator controls. The automatic pump controller shall include controls for placing the pumping assembly in three different modes of operation; manual, automatic pressure control, and automatic speed control. In addition, a variable control shall be included providing start, and variable speed/discharge ranges depending on the controller mode selection.

3.5.7.2 Hardware. The pressure sensing and speed command components shall be solid state electronics. They shall precisely position the fuel control, on the engine, isochronous (zero droop) speed settings.

3.5.7.3 Pressure Lines. The pressure lines that provide suction and discharge pressure to the automatic pump controller shall have drainage cocks for complete drainage during storage.

3.5.8 Fuel system. A fuel tank shall be furnished sufficient for not less than 8 hours of continuous operation at the rated load. The system shall be provided with a fuel gauge, graduated in 1/4 increments, minimum, from empty to full. The gauge may be located on the control and instrument panel or may be attached directly to the tank. The fuel tank shall include a hand-operated fuel shutoff valve with an arrow showing the directions of valve operation. The fuel tank outlet shall be from 0.50 inch to 1.00 inch above the bottom of the tank. The non-plastic fuel line shall be provided with a 3-way selector valve and a 0.50 inch SAE-J513 flared fitting for connection to an auxiliary fuel supply. A corrosion-resistant circular metal face-plate for the fuel selector valve and shall be marked "OFF" "UNIT TANK" and AUXILIARY". Size 2 pumping assemblies shall be provided with an additional 3-way selector valve and face-plate and two 10-foot long fuel hoses with the same fittings, with storage for the hoses shall be provided on the pumping assembly. No fuel spillage shall occur when the fuel tank is full, and the pumping assembly is operating in accordance with 3.5.2. The fuel tank shall be protected from damage by stones and other projectiles when in tow.

3.5.9 Chassis. The pumping assemblies shall be mounted on chassis having two single wheels, on an axle assembly attached to the underside of the pump frame. The axle and chassis assembly shall have a rated capacity exceeding the maximum payload, but not less than 2000 pounds for the size 1 and 6000 pounds for the size 2. The chassis shall be equipped with not less than two shock absorbers for stability.

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3.5.9.1 Tires and wheels. Tires and wheels shall be in accordance with the guidelines of the Tire and Rim Association and shall be of sufficient size and durability to support loads in off-road conditions with no more than 50 psi air pressure. Tire pressure shall be indicated directly above the tires in 1-inch high letters.

3.5.9.2 Brakes. The pumping assemblies shall be equipped with hydraulic inertia type brakes. Each pumping assembly shall be equipped with a manually operated, parking hand brake with snap-over-center locking design. The force required to apply the parking brake to keep the trailer from moving down a 25 percent slope shall not exceed 35 pounds. The parking brake handle shall be accessible from the operator's station. All cables and brake lines shall be routed so as not to interfere with the free movement of other trailer assembly components. The brake lines and all cables shall be securely clamped along the routing path and routed to avoid brush and debris damage.

3.5.9.3 Towing extension. The tongue shall have a nominal height of 26.00 inches when the unit is level and on a level surface. The tongue shall have a towing eye in accordance with SAE J847. Weight on the towing eye with the fueled pumping assembly parked on level ground shall be not greater than 15 percent and not less than 10 percent of the gross weight of the unit. Type I, size 1, and type II, size 1 pumping assemblies shall be equipped with retractable towing extensions capable of easy extension for towing and retraction for storage.

3.5.9.4 Stabilizing legs. Type I, size 1, and type II, size 1, pumping assemblies shall be equipped with an adjustable, hand-crank style jack landing gear to permit operation of the tongue by one person and two hand-crank style stabilizing legs to permit operation without tipping over when not attached to a towing vehicle. The landing gear and stabilizing legs shall each be equipped with a sand shoe with a minimum area of 150.00 square inches. Type I, size 2, and type II, size 2, pumping assemblies shall be equipped with two adjustable stabilizing legs to permit operation without tipping over when not attached to a towing vehicle. The legs shall have a minimum travel of 10.00 inches.

3.5.9.5 Lights and reflectors. The pumping assemblies shall be furnished with not less than two amber reflectors at the forward of the assembly and two red at rear. Type I, size 2, and type II, size 2, pumping assemblies shall be equipped with stoplights and taillights. The lights may be composite lights. Terminations shall be to the government furnished MS 75020 Electrical Connector (see3.8).

3.5.9.6 Battery Box. The battery box shall be made of a durable, rigid, corrosion resistant material, suitable to the application ,and shall be designed to securely retain the batteries. The batteries shall be restrained within the box by electrically insulated material and require no tools for removal. The box shall contain four 0.50 inch holes for drainage and the drain holes shall be positioned so that they drain directly to the ground. Electrolyte, as required, shall be provided and packaged as required for aircraft transport.

3.5.10 Transportability. The pumping assemblies shall be capable of being transported by military or commercial trailers, trains, marine vessels, and aircraft, and be capable of withstanding the impact forces encountered in shipment without damage or permanent deformation. The pumping unit shall be equipped with lift point, tiedown and slinging provisions.

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3.5.11 Identification marking. Each pumping assembly shall be identified on plates attached to the pumping assembly by metal fasteners in a place where the plate will be visible and legible but not interfere with the operation and maintenance of the pumping assembly. The pumping assembly shall be identified by the National Stock Number and the manufacturer's name and part number. The plate(s) shall be constructed of a durable material and shall be permanently attached with fasteners. Unless a separate transportation data plate is provided, the identification plate shall also include, as a minimum, the unit cubage and weight.

3.5.12 Cleaning, treatment and painting. All external surfaces of the pumping assembly normally painted, except those that reach a temperature of 400 °F, shall have a chemical agent resistant coating that withstands the use of common cleaners. Any external surface that reaches a temperature of 400 °F during operation shall be coated with a high temperature paint. Unless otherwise specified (see 6.2), color shall be camouflage green, color no. 34094 of FED-STD-595, for type I, size 1 and type II, size 1; and tan, color no. 33446 of FED-STD-595, for type I, size 2 and type II, size 2. All informational marking and lettering shall be flat black and caution marking in accordance with ANSI Z 535.2. Internal components may be finished with the manufacturer's standard commercial paint and color, plating or treatment.

3.6 Human Factors.

3.6.1 Safety. All shafts or rotating parts shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust components exposed to contact by personnel or that create a fire hazard, shall be fully guarded or insulated. Hot surfaces shall not be positioned near material that is flammable or that could be melted. The fuel fill port shall be located to allow refueling without the need to open any panels or doors. Fuel tanks shall be positioned to prevent spills or overflows from running onto the engine, exhaust, or electrical components. Exhaust or discharges from the equipment shall be directed away from the operator’s position. Electrical equipment shall be effectively guarded and grounded to prevent electrical hazards.

3.6.2 Noise limits. Steady-state noise produced by the pumping assemblies shall not exceed 90 dB(A) as measured in accordance with 4.5.13. If noise levels are between 85 and 90 dB(A) , or if procedures for noise suppression have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 90 dB(A) limit is obtained, hazard signs as specified by the procuring activity shall be prominently displayed on the equipment.

3.6.2.1 Noise suppression methods. Any method of noise suppression may be used as long as all components used by the operator are easily accessible without the use of any tools. The throttle control assembly shall be accessible without causing the pump assembly to exceed the required noise level while operating at full load. An enclosure shall be used as a last resort to noise suppression; it shall be easily removable, for required field maintenance, and shall include external provisions for carrying the pumping assembly. If the pumping assembly is enclosed, adequate ventilation shall be provided so that the temperature inside the enclosure does not exceed 150 °F. Any noise reducing insulation used shall be non-flammable, non-combustible, shall not absorb fuel and oil and shall be vermin proof.

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3.6.3 Maintenance and handling. The pumping assemblies shall be designed to permit easy accessibility for maintenance and service in the field. The design shall prevent conditions hazardous to personnel or deleterious to equipment. If a noise enclosure is used, any access panels or doors shall be labeled or stenciled to indicate the items accessed through them.

3.6.4 Grounding assembly for type I pumping assemblies. Two ground rods of 0.75 inch nominal diameter and 60.00 inches long shall be provided. Two uninsulated, 10 AWG, 72 inch, braided ground wires soldered to 50 ampere, uninsulated, electrical bronze, plier type clamps shall also be provided. The ground wires shall be permanently attached to the ground rods or means shall be provided for attachment, without tools, that will ensure electrical continuity.

3.7 Pumping assembly envelopes.

3.7.1 Type I, size 1, and type II, size 1. Size 1 pumping assemblies shall weigh no greater than 2800 pounds and fit within the dimensions of 87.00 inches long, 70.00 inches wide, and 77.00 inches high.

3.7.2 Type I, size 2, and type II, size 2. Size 2 pumping assemblies shall weigh no greater than 3600 pounds and fit within the dimensions of 200.00 inches long, 90.00 inches wide, and 72.00 inches high.

3.8 Government furnished property. To ensure compatibility with existing equipment, the Government will furnish one Connector, Plug, Electrical in accordance with MS 75020, for each type I, size 2, and type II, size 2 pumping assembly. The connector shall be wired as specified below as applicable.

TERMINAL	TERMINATION
A	LH BLACKOUT MARKER
B	LH SERVICE SIGNAL
C	RH BLACKOUT MARKER
D	GROUND
E	SERVICE CLEARANCE
F	BLACKOUT STOP
H	BLACKOUT CLEARANCE
J	RH SERVICE SIGNAL
K	AUXILIARY POWER
L	GROUND
M	BRAKE
N	BRAKE

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article inspection is required, it shall be performed on one complete pumping assembly. The inspection shall include the examination of 4.4 and the testing of 4.5.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4.

4.4 Examination. Each pumping assembly shall be physically examined for compliance with the requirements specified in 3.2 through 3.7. Any redesign or modification to comply with specified requirements, or any redesign or modification necessary following failure to meet the specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of performance and dimensional requirements. Non-compliance with any specified requirement, or the presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Testing.

4.5.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with the applicable provisions of the test code for centrifugal pumps of the standards of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions of a barometric pressure of 29.92 inches of mercury and water 60 °F. Water at a temperature of between 60 and 80 °F may be used as the test fluid.

4.5.2 High temperature. The pumping assembly shall be stored at a temperature of 160 °F; except type I, size 1, and type II, size 1, that are 155 °F; for 24 hours and then the temperature lowered to 135 °F; except type I, size 1, and type II, size 1, that are 120 °F. After temperature stabilization, at 135 °F or 120 °F, the pumping assembly shall be started and operated at rated capacity for 3 hours. Nonconformance to 3.3.1 and 3.3.2 shall constitute failure of this test.

4.5.3 Low temperature. The pumping assembly shall be stored at -65 °F for 24 hours. For a type I pumping assembly, raise the temperature to -25 °F and stabilize, then start and operate the pumping assembly at rated capacity for 3 hours. For a type II pumping assembly, raise the temperature to 32 °F and stabilize, then start and operate the pumping assembly at rated capacity for 3 hours. Nonconformance to 3.3.1 and 3.3.2 shall constitute failure of this test.

4.5.4 Drop. The pumping assembly, balanced on only its own two wheels, shall be allowed to fall forward onto the pintle hook and backward until it contacts the ground. The assembly shall then be inspected for damage or broken components and operated for not less than one half hour. Any breakage, permanent deformation, or failure of the pumping assembly to operate as specified in 3.5.2 shall constitute failure of this test. Nicks, dents, scrapes, or a marred finish shall not be considered defects.

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4.5.5 Shock. The pumping assembly shall be subjected to a test to determine its capability of withstanding the shocks normally induced by transport, loading, and unloading in the field. The unpackaged pumping assembly shall be anchored to a domestic service railway freight car and shunted at speeds of not less than 4, 6, and 8 mph ,respectively, against two stationary flatcars having a total gross weight of not less than 240,000 pounds with their air brakes set in the emergency application position and draft gear slack removed. The pumping assembly shall face in the direction of the two cars. Repeat the procedure at not less than 8 mph with the rear of the pumping assembly in the direction of the two cars. Once the test has begun , there shall be no adjustment of the tiedowns or other anchoring material except as necessary to reverse the direction of the pumping assembly. After each impact the pumping assembly shall be inspected. After completion of all impact tests, an operational checkout of the pumping assembly shall be conducted. Any breakage, permanent deformation, or failure of the pumping assembly to operate as specified in 3.5.2 shall constitute failure of this test.

4.5.6 Transportability. Each type I, size 1 and type II, size 1 first article test unit shall be subjected to the following subordinate testing. At the end of each phase, the unit shall be operated for not less than one half hour. Negatively affecting the tow vehicle's stability during testing, any detrimental loosening, permanent deformation, or damage to any component or ancillary item of the pumping assembly, or failure of the unit to meet the performance requirements of 3.5.2, shall constitute failure of this test.

4.5.6.1 Fording. The unit, in its transport mode, shall be towed through water at a depth of 30.00 inches, minimum, at a speed not greater than 8 mph. The unit shall be operated within five minutes of extraction from the water and inspected for damage.

4.5.6.2 Slope. The unit, in its transport mode, shall be towed up-grade, then down-grade, on a ramp of not less than a 30 percent slope, in both forward and reverse directions. During each negotiation the unit shall be observed for contact with the surface, of other than tires. The unit shall then be towed, transversely, along a side slope of not less than 20 percent, at a speed not greater than 8 mph. The procedure shall be repeated in the opposite direction.

4.5.6.3 Mobility. The unit, in its transport mode, shall undergo the cycles of transportation tests defined in table I. The testing on different surfaces need not be continuous but each test shall be uninterrupted. During testing on improved (primary or secondary paved) roads, the unit shall be subjected to emergency stops moving forward at 20, 30, and 45 mph, and in reverse at 10 mph, five consecutive times at each speed. During cross-country testing, the unit shall be subjected to a 360 degree turn within a radius of 50 feet. A visual inspection shall be conducted after every cycle, to determine whether testing should continue.

TABLE I. Transportation survivability tests

Surface	Speed (max)		Distance (min)
	Size 1	Size 2	
Improved (paved) roads	20 mph	50 mph	100 miles
Unimproved (unpaved) roads	10 mph	20 mph	100 miles
Cross-country	8 mph	8 mph	50 miles

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4.5.7 Maintenance ratio. The maintenance ratio shall be computed during first article testing. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing. Nonconformance to 3.4.1 shall constitute failure of this test.

4.5.8 Reliability. The first article pump assembly shall be operated at its rated capacity under the standard conditions specified in 4.5.1. The pump assembly shall be continuously operated for not less than 9 hours out of every 24 hours. A failure is defined as any malfunction which:

a. Cannot be corrected within 30 minutes by adjustment, repair or replacement using only the maintenance tools and repair parts furnished with the equipment; or
b. May cause failure to commence operation, cessation of operation, or degradation of performance below specified level; or
c. May damage pump assembly by continued operation; or
d. May cause a safety hazard to operating personnel.

The total test time is total unit hours of pump "on" time and is expressed as multiples of specified MTBF (see 3.4.3). Normal maintenance shall be performed in accordance with the manufacturer's operation and maintenance schedule. Nonconformance to 3.4.3 shall constitute failure of this test.

4.5.9 Hydrostatic pressure. The non-operating pump shall be subjected to a hydrostatic pressure test, at 200 psi, for a period of not less than 5 minutes. Gauges may be excluded from testing. Any leakage, deformation, or failure of components shall constitute failure of this test.

4.5.10 Pumping assembly (operational). Test the pumping assembly for the requirements specified in 3.5.2 in accordance with ANSI/HI 1.1-1.6. Data relative to head and capacity of the pump, and speed (rpm) of the engine shall be measured at not less than six points on the pump performance curve. The measured points shall include the rating points (see 3.5.2), wide-open discharge, and shutoff. The pump shall run for not less than 5 minutes at each measured point. Delivery of less than the capacities specified in 3.5.2 or failure of the engine to meet any of the operational requirements or conditions specified in 3.5.6 shall constitute failure of this test.

4.5.10.1 Priming. Fill the pump case with water at conditions as specified in 4.5.1 and conduct the priming test with a static suction lift equivalent to 10 feet of water using the specified size suction hose. Inability of the pump to prime and deliver full capacity in 2 minutes shall constitute failure of this test.

4.5.11 Safety. The pumping assembly shall be evaluated for hazards throughout testing. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.12 Human engineering. Human engineering requirements shall be evaluated throughout testing. Nonconformance to 3.6.3 shall constitute failure of this test.

4.5.13 Noise level test. Noise levels shall be measured with the pumping assembly operating under full load, and at not fewer than 12 points around the equipment including the operator’s and occasionally occupied positions. The operator's position is defined as 24.00 inches horizontally and 12.00 inches vertically from the throttle control. Occasionally occupied positions are defined as 39.00 inches from the equipment, 55.00 inches above the ground. Noise levels shall be provided as a dB(A) level. Failure to comply with 3.6.2 shall constitute failure of this test.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type I, size 1, pumping assemblies are for use in transfer and dispensing of potable water. Type I, size 2, pumping assemblies are components of the Inland Petroleum Distribution System (IPDS) and transfer petroleum products to and from large temporary storage facilities. Type II, size 1, pumping assemblies are a component of the Fuel System, Supply Point: Portable, 60,000 and 120,000 Gallon and are intended for transfer and dispensing of liquid fuel. Type II, size 2, pumping assemblies are components of the Tactical Water Distribution System (TWDS) and are intended to pump potable water. The pumping assemblies covered under this specification are military unique because they must operate under extreme environmental conditions (see 3.3) under which commercial items would experience catastrophic failure.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this publication.
b. Type, size and style of pump required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
d. When a first article inspection is required, the number of units required, and the time frame for submission (see 3.1).
e. Color when other than as specified (see 3.5.12).
f. Packaging required (see 5.1).

6.3 Data Requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractors maintenance and operation manuals to be furnished with each pumping assembly.

6.4 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for specification MIL-PRF-53051.

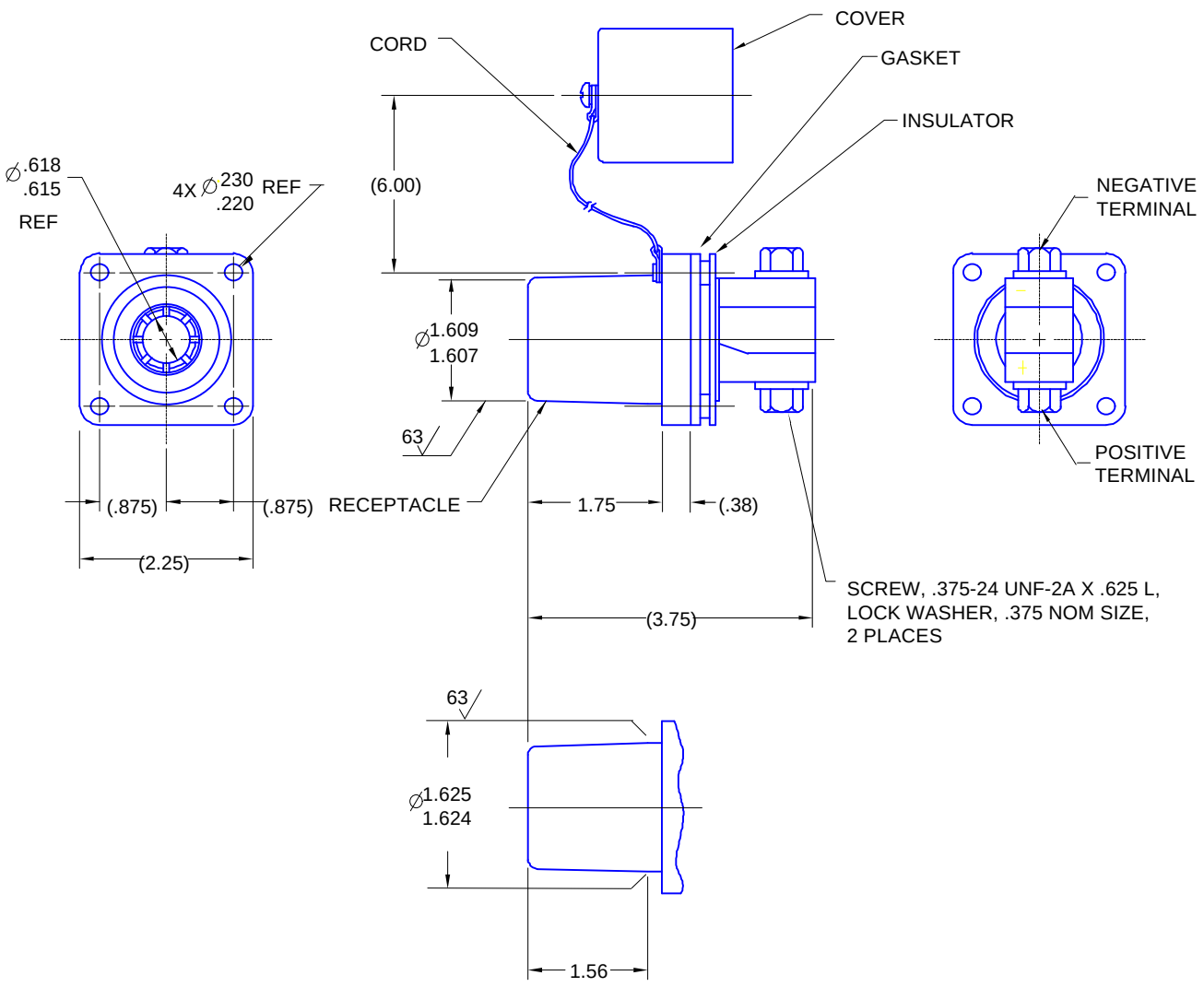
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M53051-2-2-2
- 2 - Without speed/pressure regulator (style B)
- 2 - 600 GPM capacity at 350 feet total head (size 2)
- 2 - Pumping assembly , water (type II)

6.6 Subject term (key word) listing.

Pump, Centrifugal, Self-Priming
Pump, Liquid Bulk Transfer
Pump, Fuel
Pump, Water
Pump, 350 GPM
Pump, 600 GPM

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NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
   TOLERANCES: .XX ± .06.

Figure 1. Auxiliary Charging Receptacle.

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Custodians:
Army - AT
Air Force – 99
Navy –SH

Review activity:
DLA - CC

Preparing activity:
Army – AT

(Project 4320-0046)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter must be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-53051B	2. DOCUMENT DATE (YYMMDD) 980806

3. DOCUMENT TITLE
Pumping Assemblies, Wheel Mounted, Diesel Engine Driven, Bulk Transfer, Fuel and Potable Water Pumping Service

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME USA TACOM, AMSTA-TR-D/210	b. TELEPHONE (Including Area Code) (1) Commercial (810)574-7744 (2) AUTOVON DSN 786-7744
C. ADDRESS (Include Zip Code) U.S. Army Tank-Automotive and Armaments Command Attn: AMSTA-TR-D/210 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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